Query Performance Prediction Is Not Benefit Prediction: Observability for Selective Retrieval Interventions

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Abstract

Query performance prediction (QPP) estimates whether a ranking is poor, but selective retrieval needs a different prediction: whether an intervention will improve it. We formalize this distinction as a decision under observable telemetry and derive an exact, cost-aware decomposition of the gain that no policy intervening only on failing queries can reach. We evaluate LLM query expansion, cross-encoder reranking, and pseudo-relevance feedback on four benchmarks, two of them pre-registered and untouched. Without a budget cap, a perfect failure gate recovers only 8–57% of the oracle's gain on TripClick and NQ (18–46% on ArguAna and FiQA). On the same queries, failure detectors identify retrieval failure but predict intervention benefit weakly; on pre-registered FiQA the AUCs are 0.72 and 0.46–0.52. Budgeted evaluation reveals a second limit: learned failure gates spend much of their budget on failing queries the intervention does not rescue, even where the restriction loosens. Stronger pre-action predictors do not close the gap, and partial probes of the intervention improve routing on NQ but not on TripClick, ArguAna, or pre-registered FiQA. Selective retrieval should be evaluated by intervention benefit, oracle headroom, and cost-aware routing, not by failure-prediction accuracy alone.

1 Introduction

Retrieval interventions help some queries and harm others. LLM query expansion (QE) can rewrite a query using the evidence it retrieves [14], yet it harms a measurable fraction of queries [1, 27]; cross-encoder reranking and pseudo-relevance feedback (PRF) are equally uneven. The natural response is to intervene selectively, and the natural gate is a query performance predictor (QPP): intervene when the current ranking looks like it is failing [2, 10]. That the gate's target differs from the quantity a policy needs is not new: selective-QE work predicts expansion benefit directly [6], and uplift modelling shows that targeting high-risk cases is not targeting the cases a treatment helps [3, 8]. What is missing is how much a failure gate gives up, why, and whether the telemetry a deployed system can observe, its *observability*, can do better.

Contributions. (C1) A decision framing of selective intervention under observable telemetry, with an exact decomposition of the gain no policy acting only on failing queries can capture, at zero and positive cost (Section 3). **(C2)** Evidence on four corpora, two pre-registered and untouched, that the targets differ and that the difference has measurable consequences: the uncapped *failure-only ceiling* is 8–57% of the oracle's gain, failure scores rank helped queries near chance, and a learned failure gate also selects poorly among the failing queries it may target. **(C3)** Evidence that better information is not automatically enough: at matched budgets and measured cost, stronger pre-action benefit predictors and partial probes of the intervention do not consistently improve routing.

2 Related Work

QPP and selective QE. Pre- and post-retrieval QPP (Clarity [5], WIG [30], NQC [21]) predicts query difficulty. Cronen-Townsend et al. [6] compare the clarity of expanded and unexpanded rankings; Amati et al. [2] gate expansion on difficulty predictors; Lv and Zhai [15] predict a feedback weight; Collins-Thompson [4] trades reward against risk. Even idealised selective QE needs QPP at Kendall $\tau \geq 0.5$ [9]. Such work sizes predictor accuracy or evaluates gated systems; none bounds what a failure gate can recover or compares gates at matched budgets.

LLM QE and adaptive retrieval. Query2doc [26], MuGI [29], and CSQE [14] expand unconditionally; generative expansions often hurt strong retrievers [27] and fail on unfamiliar or ambiguous queries [1]. Kotte [11] gates prompt-only rewriting for dense retrievers without reliable gain. Sarode [19] finds that a learned selector over candidate LLM rewrites captures little oracle headroom and transfers inconsistently; we ask whether to intervene at all, across three interventions, and bound failure-restricted policies. Adaptive-RAG [10] routes on question complexity and Mallen et al. [17] retrieve only for less popular entities; cascade rankers [25] spend costly stages only on promising candidates, and cutoff and efficiency predictors [7, 23] set per-query reranking depth or pruning. In uplift terms [8], both outcomes are observed offline, so our label is exact up to relevance noise.

3 Framework

For a query q, let R_0 be the original ranking and R_a the ranking after intervention a, with metric M (nDCG@10). The intervention's effect is $\Delta_a(q) = M(R_a, q) - M(R_0, q)$ and its cost c_a ($\Delta_0 = c_0 = 0$). With full information the best decision is $a^*(q) = \arg\max_a\{\Delta_a(q) - \kappa c_a\}$, for a cost weight κ. A deployed controller sees only telemetry Z, so the best it can do is $\pi^*(Z) = \arg\max_a\{\mathbb{E}[\Delta_a \mid Z] - \kappa c_a\}$. What matters is therefore what Z reveals about Δ_a, not about $M(R_0)$. Two information sets differ in cost: *pre-action* telemetry Z_0 describes R_0 and costs no intervention, though computing it is not free; *post-action* telemetry Z_1 also describes R_a and exists only after c_a is paid. A *probe* observes part of the intervention, such as cross-encoder scores of a few candidates, at part of its cost (Section 9). A failure gate replaces $\mathbb{E}[\Delta_a \mid Z_0]$ with $P(F \mid Z_0)$, where $F = \mathbf{1}[M(R_0, q) = 0]$.

The failure-only ceiling. Fix one intervention with $c_a = 1$, so κ is its cost per query in nDCG@10 units. The oracle selector earns $G_{\text{orc}}(\kappa) = \mathbb{E}[(\Delta - \kappa)_+]$. A policy that may act only on failing queries does best by intervening where $F = 1$ and $\Delta > \kappa$, earning $G_F^*(\kappa) = \mathbb{E}[F(\Delta - \kappa)_+]$ (the *failure-restricted oracle*; its share of G_{orc} is the failure-only ceiling), so

$$G_{\text{orc}}(\kappa) - G_F^*(\kappa) = \mathbb{E}[(1 - F)(\Delta - \kappa)_+]$$

(gain on healthy queries) bounds every failure-restricted policy at any $\kappa \geq 0$. Because $M \geq 0$, a failing query cannot be harmed

($\Delta \geq 0$ when $F = 1$), so at $\kappa = 0$ a *perfect* failure gate, intervening iff $F = 1$, attains G_F^*. At $\kappa > 0$ it earns $G_F(\kappa) = \mathbb{E}[F(\Delta - \kappa)]$, a further $\mathbb{E}[F(\kappa - \Delta)_+]$ below G_F^* spent on failing queries helped by less than κ, which only information beyond the binary label avoids. The identity is elementary; its use is that every term is measurable offline for any intervention. It does not bound a policy built from a failure *score* that also acts on healthy queries.

4 Setup

Data. TripClick [18]: 1.52M health documents, the TAIL bucket's validation and test splits (1,175 queries each), with click-derived relevance labels. BEIR NQ [12, 22]: 2.68M passages, human labels; 800 validation and a disjoint 2,052-query test set. BEIR ArguAna [24]: 8,674 arguments, 1,406 queries whose relevant document is the counterargument; BEIR FiQA [16]: 57,638 financial posts, official splits (Section 8).

Interventions. *LLM QE*: BM25 over SQLite FTS5; a deterministic rule picks one of four expansion operators or none, and `qwen2.5:3b` (4-bit, greedy, ≤ 2 calls) writes a corpus-grounded expansion. On TripClick and under dense retrieval (`bge-small-en-v1.5` [28]) the expanded ranking's top candidates are reranked by a MiniLM cross-encoder [20]; the original is not, so those labels include the reranker's effect. TripClick's deployed QE fuses original and expanded rankings (RRF); NQ's uses the expanded ranking. *Reranking*: the cross-encoder scores the BM25 top 100; its top 50 lead. *PRF*: RM3-style [13] weights from the top d documents select t terms whose BM25 ranking is interpolated with the original (weight λ); (d, t, λ) is tuned on validation over 108 settings.

Labels and models. *Failure*: $F = 1$ (45% of TripClick and 49% of NQ test queries under BM25). *Harmed*: $\Delta < 0$, ties not harmed. Detection is a logistic regression on 8 pre-action features (score shape, term coverage, rare-term IDF, entity overlap, coherence, perturbation stability; 6 under dense). Action utility is fit on Δ from post-action features. For pre-action routing, a *benefit* model regresses Δ on the 8 pre-action features alone. Every model and threshold is fit on validation and scored once on test; intervals are paired bootstraps over queries (10^4 resamples). We call the deployed detector, used to route, the *learned failure gate*, and any model of Δ a *benefit predictor*.

Evidence ledger. Only the ArguAna and FiQA results (Section 8) are independent confirmation: every NQ query and the TripClick TAIL test set have been inspected. *Fixed before a test run*: the NQ, reranking, and PRF runs. *Pre-committed analyses of inspected traces*: the AUCs, ceiling, budget routing, and Holm correction. *Exploratory*: TripClick test was used during development, the changed-query harm comparisons were added after seeing the main results, and the NQ B1/B2 runs, diagnostics, and probes of Section 9 followed protocols frozen before computing but after the test sets were inspected; the NQC probe contrasts, Spearman intervals, the positive-cost and budget-capped ceilings, same-target and within-failing AUCs, and P95 costing were added after review. Precommitments and scripts are in the anonymized repository: https://anonymous.4open.science/r/agentic-query-expansion-8939/.

Table 1: Failure-only ceiling (test). Rescue: $P(\Delta > 0 \mid F)$. Healthy helped / harmed: $P(\Delta \gtrless 0 \mid \neg F)$. Share: perfect failure gate's gain over the oracle's at $\kappa = 0$, 95% CI. ArguAna, FiQA: pre-registered, untouched test (n = 430, 648).

Setting	$P(F)$	Rescue	Healthy help	harm	Share
LLM QE, TripClick	.45	.07	.15	.19	42% [32, 52]
LLM QE, NQ	.49	.07	.19	.33	33% [26, 40]
LLM QE, TripClick, dense	.59	.06	.25	.41	34% [25, 43]
LLM QE, NQ, dense	.26	.15	.19	.29	33% [27, 40]
Reranking, TripClick	.45	.17	.24	.56	52% [44, 60]
Reranking, NQ	.49	.46	.55	.19	57% [54, 60]
PRF, TripClick	.45	.01	.12	.20	8% [1, 17]
PRF, NQ	.49	.05	.20	.22	32% [25, 39]
Reranking, ArguAna	.21	.36	.25	.53	38% [27, 49]
PRF, ArguAna	.21	.01	.05	.06	18% [0, 52]
LLM QE, ArguAna	.21	.27	.23	.59	39% [27, 52]
Reranking, FiQA	.51	.29	.52	.23	46% [39, 53]
PRF, FiQA	.51	.02	.14	.19	28% [11, 46]

5 The Failure-Only Ceiling

Failure-restricted policies miss most of the gain. The failure-restricted oracle captures 8–57% of the oracle's gain (Table 1; 18–46% on ArguAna and FiQA), though it never makes a query worse. The share follows how often an intervention rescues a failing query relative to how often it helps a healthy one: reranking rescues 17–46% of failing queries, so up to half its headroom lies there; LLM QE rescues 6–15% and PRF almost never. The rest sits in healthy queries the intervention also often harms, which only benefit prediction can separate. At a per-query cost of 0.02 the shares are similar (8–57%; 22–47% on ArguAna and FiQA), and intervening on every failing query then loses outright for PRF and dense TripClick QE.

A budget cap loosens the restriction only where rescues are common (exploratory, post-review). At a 10% cap, failing queries hold 98% [97, 99] of the capped oracle's gain for NQ reranking and 54–67% for the other reranking and ArguAna QE settings; the other seven stay near their uncapped shares. From a 25% cap only NQ reranking (80%) stays above its uncapped share. Pre-registered FiQA agrees: reranking rises from 46% to 83% at 10%, PRF stays at 28%.

The learned failure gate also selects poorly. At a 10% budget, 35–80% of its picks are failing queries, yet it realises at most 34% of the capped failure-restricted oracle's gain, and its gain is negative in five settings (all three on ArguAna) and in 7–9 of 11 at 25–75%. Among failing queries its score ranks the rescued ones no better than chance (Table 2; below it in four settings, not on FiQA). At a 25% budget, 45–68% of its picks on TripClick and NQ (30–36% on ArguAna) are failing queries the intervention leaves unchanged (exploratory).

The ceiling and the selection error are separate constraints. Restricting action to failing queries excludes improvements on healthy queries; imperfect prediction additionally spends budget on failures the intervention does not rescue. Both operate at every budget below 100%.

6 Detection vs. Benefit

Failure detection is ordinary QPP; benefit prediction is not. The learned failure gate ranks failing queries at AUC 0.64–0.69 (Table 2), as NQC [21] does. Scored against whether the intervention

Table 2: One score, two targets (test). Failure: AUC of the learned failure gate's score for F. Helped: its AUC for $\Delta > 0$ on the same queries, 95% CI. Within failing: helped AUC among failing queries (–: fewer than 20 rescued). Exploratory except FiQA (pre-registered).

Setting	Failure	Helped	Within failing
LLM QE, TripClick	0.65	0.52 [0.46, 0.57]	0.47
LLM QE, NQ	0.68	0.50 [0.46, 0.53]	0.44
LLM QE, TripClick, dense	0.64	0.50 [0.45, 0.54]	0.51
LLM QE, NQ, dense	0.69	0.56 [0.53, 0.59]	0.50
Reranking, TripClick	0.65	0.47 [0.43, 0.51]	0.44
Reranking, NQ	0.68	0.48 [0.46, 0.51]	0.37
PRF, TripClick	0.65	0.49 [0.42, 0.55]	–
PRF, NQ	0.68	0.45 [0.41, 0.48]	0.34
Reranking, ArguAna	0.65	0.43 [0.37, 0.49]	0.34
PRF, ArguAna	0.65	0.54 [0.39, 0.68]	–
LLM QE, ArguAna	0.65	0.42 [0.36, 0.48]	0.36
Reranking, FiQA	0.72	0.52 [0.47, 0.57]	0.56
PRF, FiQA	0.72	0.46 [0.37, 0.54]	–

helps (exploratory, post-review), the same scores reach only 0.42–0.56, with intervals below 0.5 for NQ PRF and ArguAna QE and reranking and above only for dense NQ QE. Pre-registered on FiQA, the gap replicates: 0.72 against 0.52 (reranking) and 0.46 (PRF), both Holm $p < 0.001$.

Benefit predictors do better but are weak (helped AUC $\leq$ 0.58; NQ probe 0.65). They beat the failure score on NQ reranking (+0.08 to +0.17, probe largest), NQ PRF, and ArguAna QE and reranking (+0.07 to +0.12); elsewhere differences span zero, except that the linear model trails it on TripClick PRF (−0.06).

The detector's lead over harm prediction does not contradict this. Detection beats harm AUC by +0.02 to +0.33 (Holm-significant in seven of eight original settings; +0.22 and +0.20 on FiQA), but failing queries cannot be harmed and are mostly unchanged, so the harm label is largely a failure indicator; on changed queries the gap vanishes in every TripClick and NQ BM25 setting (table in the repository).

7 Routing at Matched Budgets

Pre-action routing. At every budget, each scorer that decides before intervening stays close to random routing and far below the oracle (Table 3). Failure routing beats random by at most 0.006 at any budget and often falls below it (Figure 1), and at a 25% budget each pre-action scorer's picks are helped within −0.11 to +0.08 of the base rate. Averaged over budgets, NQC is no better than the detector, and benefit routing beats failure routing only for reranking on NQ (+0.009 [+0.003, +0.015]), where it leads at every budget (per-budget CIs in the repository). Boosted and embedding predictors (B1, B2; Section 8), fit on NQ validation (exploratory), do no better: none survives Holm, and the best recovers 8% of the oracle-minus-failure headroom. Failure routing can still save cost: for LLM QE, the deployed failure-selective policy halves generator calls and on NQ recovers most of unconditional QE's loss (+0.015 [+0.008, +0.022]); on TripClick no QE policy differs from BM25.

8 Pre-Registered Replications

ArguAna. *Question:* do stronger pre-action benefit predictors beat failure routing on data never inspected for this work? *Design*, pre-registered before download (`configs/arguana_precommit.json`): a debate-grouped split (701/275/430), the three interventions (LLM QE as on NQ), and Holm-corrected primary contrasts for three new predictors on 10 pre-action features (the 8 above, NQC, query length): B1, LightGBM regression on Δ; B2, distance-weighted mean Δ of the k nearest training queries under `bge-small-en-v1.5`; B3, LightGBM quantile regression (its registered quantile choice is unsound, so a secondary B3 fixes 0.25). Leakage fixes (each query is a corpus document; 108 arguments are cross-listed) and three externally proposed contrasts were dated amendments made before any label was read. *Result: no.* Every intervention loses on average (reranking −0.071 [−0.108, −0.034], LLM QE −0.163), yet the oracle could gain 0.053 and 0.010 budget-averaged, so the task is pure selection. For reranking, B1 minus failure routing is +0.015 [−0.001, +0.028] (Holm $p = 0.13$), B2 minus failure +0.008 (Holm $p = 0.37$), B3 minus B1 under a budget cap +0.006 ($p = 0.52$); the best recovers 16% of the oracle-minus-failure headroom, below the pre-registered 25% bar, and a secondary family agrees (none of nine survives Holm). Failure routing of reranking is below random at every budget (Figure 1). PRF changes only 9% of rankings, so its rows are uninformative. At 430 queries, B1's +0.015 may be real but is unresolved.

FiQA. *Predictions*, fixed before download in `configs/fiqa_precommit.json`: (P1) the failure score ranks failure better than benefit; (P2) among failing queries it ranks the rescued ones below chance; (P3, P4) since reranking helps on validation (+0.10), Probe-10 routing beats failure and B1 routing. Reranking and PRF follow the ArguAna protocol (1,500 training, 500 dev, 648 test queries; one Holm family, five evaluable tests). *Confirmed:* failure detection does not translate into benefit prediction (P1: 0.72 against 0.52 and 0.46, Holm $p < 0.001$). *Not confirmed:* the within-failing inversion (0.56 [0.49, 0.63]; PRF, with 8 rescues, not evaluable) and the probe (minus failure −0.001 [−0.009, +0.008], minus B1 +0.006 [−0.001, +0.013]; Holm $p \geq 0.25$). B1 and B2 do not beat failure routing either (−0.006 each, Holm $p = 0.19$).

9 Partial Observation and Cost

A probe sits between pre- and post-action telemetry. In an exploratory analysis (protocol frozen before computing), we score only the first $k \in \{5, 10, 20\}$ BM25 candidates ($k = 10$ primary, fixed in the protocol) with the cross-encoder, regress the Δ of full reranking on nine probe features with B1's model and selection rule, and fully rerank the top-scoring queries, reusing the probe's scores. Each policy is charged its median measured latency (50 validation queries per corpus; Figure 1): full reranking takes 0.48, 1.11, 1.26, and 0.84 s on NQ, TripClick, ArguAna, and FiQA, a ten-candidate probe 0.05–0.13 s.

On NQ, where reranking usually helps (Figure 1), Probe-10 routing beats failure routing by +0.024 [+0.017, +0.031] and B1 by +0.017 [+0.011, +0.022] over 10–75% reranked (both Holm-significant over the six primary tests), and its predictions rank Δ best (Spearman 0.25 [0.21, 0.29]; B1 0.09, NQC −0.06). NQ's pre-action telemetry (a perturbation search over 2.68M passages)

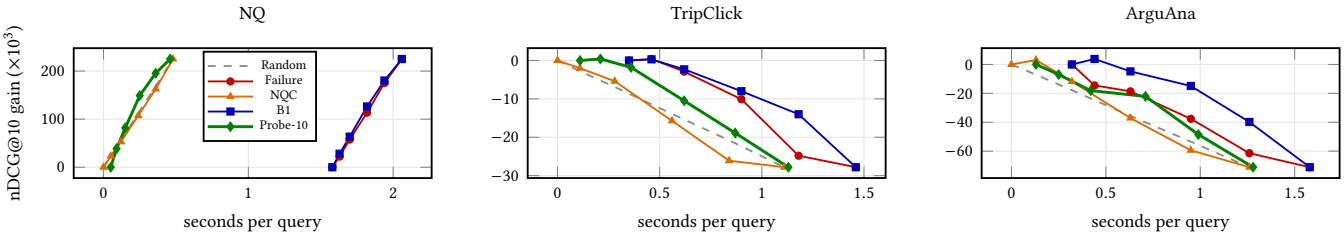


Figure 1: Reranking gain over BM25 against measured cost per query (exploratory) at 0–100% of queries reranked; cost is telemetry (Failure, B1, paid before deciding) or the probe, plus reranking. Oracle (off scale): 0.25, 0.06, 0.10.

Table 3: Pre-action routing gain (nDCG@10, $\times 10^3$) by share intervened on: failure (F), benefit (B), random (R, expected). Oracle and B−F [95% CI]: means over the five budgets (registered; Oracle can fall below All); * Holm-significant over the six settings.

	10%			25%			50%			75%			100%		
Setting	F	B	R	F	B	R	F	B	R	F	B	R	All	Oracle	B−F (avg.)
LLM QE, TripClick	3.5	1.2	0.1	3.0	−0.8	0.2	3.0	0.8	0.3	−1.5	1.0	0.5	0.7	19.7	−1.2 [−3.6, +1.4]
LLM QE, NQ	1.2	0.8	−2.2	−2.2	−1.2	−5.4	−5.9	−2.8	−10.8	−11.6	−12.6	−16.3	−21.7	28.4	+0.6 [−2.0, +2.9]
Rerank, TripClick	0.3	0.3	−2.8	−3.1	−0.8	−6.9	−10.1	−8.6	−13.9	−24.8	−19.7	−20.8	−27.8	40.5	+1.8 [−3.7, +6.3]
Rerank, NQ	22.1	32.6	22.5	57.6	66.9	56.3	113.5	129.2	112.5	175.1	183.7	168.8	225.1	199.5	+8.8 [+3.3, +15.0]*
PRF, TripClick	−0.7	0.3	−0.1	−0.2	0.7	−0.3	0.5	−0.2	−0.6	−0.5	−1.1	−0.8	−1.1	9.6	+0.1 [−1.0, +1.0]
PRF, NQ	0.1	0.2	1.0	1.4	1.2	2.4	3.9	5.7	4.9	7.5	7.6	7.3	9.7	24.0	+0.4 [−1.7, +2.4]

takes longer than reranking every query, so always reranking dominates failure and B1 routing. Against free NQC (post-review), probe routing gains +0.030 [+0.022, +0.037] at matched fraction and +0.020 [+0.012, +0.028] at equal median-latency cost (both Holm-significant over six NQC contrasts), +0.010 [+0.002, +0.018] at P95. On TripClick and ArguAna, where reranking loses on average, probes do not help: no matched-fraction contrast is significant, and B1 ranks Δ at least as well (Spearman 0.08 and 0.10 vs. 0.05; differences span zero); nor on FiQA, where it helps (Section 8).

Robustness (secondary, uncorrected). Every probe size beats failure and B1 on NQ; none does so on TripClick or ArguAna.

10 Discussion and Limitations

QPP and selective retrieval predict different targets, and the difference matters. One consequence is proven: at any cost, failure-restricted policies are capped by the oracle gain on failing queries (8–57% uncapped). The other is empirical, for these representations, models, and corpora: failure scores rank helped queries near chance, and the learned failure gate spends much of its budget where the intervention does nothing. The second alone would fit 'no pre-action signal we test predicts benefit'; the first is what makes failure specifically the wrong target. Better information is not automatically enough: stronger pre-action predictors and partial probes improved routing only on NQ. For these systems only, always rerank on NQ and FiQA when affordable and skip it on TripClick and ArguAna.

Limitations. One small (3B) generator; reranking and PRF only under BM25, with one cross-encoder and one PRF family. Four corpora, two untouched (430 and 648 test queries), underpowered for differences below about 0.015; under dense retrieval, retriever strength and domain are confounded, and the dense QE labels include the reranker. TripClick's labels are click-derived and sparse. The cost weight κ is illustrative, latencies are single-machine (median, P95 check), and probes cover reranking only. All pre-committed results are reported.

11 Conclusion

QPP predicts whether a ranking is poor; selective retrieval needs to know whether an intervention helps. A controller acting only on failing queries gets at most 8–57% of the oracle's gain, a learned failure gate selects poorly even among them, and its score ranks helped queries at AUC 0.42–0.56, replicated on untouched FiQA. Selective-intervention work should report benefit AUC, failure-only ceilings, and cost-aware routing curves, not only failure-detection AUC.

Ethical Considerations

All data (TripClick, BEIR NQ, ArguAna, and FiQA) are existing research benchmarks used under their licenses; TripClick is used for non-commercial research, and no model or per-query output fit on it is released. TripClick's click logs are aggregate, de-identified interactions, and we make no attempt to re-identify searchers. No human-subjects data was collected. The generator is a small, locally hosted model used only to rewrite queries within a bounded action space. Our central finding cautions against deploying generative interventions on the strength of a failure detector alone.

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